

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: ASAP++ | Context: Qatari and Arab High School Arabic Essay Writers (Grades 10–12)

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark output ontology is a fixed integer score range **[Q11]** over a small set of attributes (content, organization, word choice, sentence fluency, conventions for narrative; analogous traits for other types) **[Q16, Q17, Q18, Q19]**, with annotations described as gold standard for predicting attribute scores **[Q50]**. The deployment requires natural-language revision suggestions with explanatory rationale (per elicitation Q3/A3), which is a categorically different output space — not a numeric score taxonomy at all. Additionally, traits like 'word choice' and 'sentence fluency' are calibrated to English vocabulary range and syntactic variety; Arabic writing quality dimensions critical for this deployment include morphological correctness, classical rhetorical structure (ةحاصف / ةغالاب), and dialect intrusion handling [cultural_norms_notes, writing_systems.morphological_considerations], which are not represented.

ID	Evidence
Q3	"The ASAP AEG dataset comprises of approximately 13,000 essays, written across 8 prompts. The essays were written by students of class 7 to 10. Each essay was evaluated by 2 evaluators. 6 out of the 8 prompts only have overall scores. Only 2 of them have scores for individual essay attributes, like content, organization, style, etc." (p.1)
Q11	"We use the same score range as the overall score range of the essays." (p.2)
Q16	"The ASAP dataset already contains attribute scores for the narrative essays, namely content, organization, word choice, sentence fluency, conventions, etc." (p.2)
Q17	"Since we already have scores present for the narrative essays, we describe the scores for the other types of essays." (p.2)
Q18	"Based on the types of essays, there are 2 sets of attributes." (p.2)
Q19	"There are 5 attributes for narrative essays, namely 1. Content: The quantity of relevant text present in the essay. 2. Organization: The way the essay is structured. 3. Word Choice: The choice and aptness of the vocabulary used in the essay." (p.2)
Q50	"These annotations can be used as a gold standard for future experiments in predicting different attribute scores." (p.5)

Strengths

- The trait inventory (content, organization, vocabulary/word choice, fluency, conventions) **[Q19]** partially overlaps with the seven-trait rubric used in LAILA **[WEB-12]** (relevance, organization, vocabulary, style, development, mechanics, grammar), so the methodological design is a useful precedent for an Arabic rubric — even if the labels themselves are not transferable.

ID	Evidence
Q19	"There are 5 attributes for narrative essays, namely 1. Content: The quantity of relevant text present in the essay. 2. Organization: The way the essay is structured. 3. Word Choice: The choice and aptness of the vocabulary used in the essay." (p.2)
WEB-12	LAILA used annotators trained on Arab high school writing — example of appropriate annotator profile for this deployment (https://arxiv.org/abs/2512.24235)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: The benchmark's label categories [Q19] are partially conceptually relevant (content, organization) but miss categories central to MSA writing (morphological correctness, rhetorical eloquence, dialect intrusion) [writing_systems, cultural_norms_notes].

OO-2: Missing categories include: morphological/grammatical correctness as a distinct trait, adherence to classical Arabic rhetorical norms (ةغالـب), diacritical accuracy where applicable, and dialect-intrusion handling [writing_systems.morphological_considerations, languages.diglossia_note].

OO-3: Yes — 'word choice' and 'sentence fluency' [Q19] encode English vocabulary-range and syntactic-variety assumptions that map poorly onto MSA writing-quality criteria.

OO-4: Stakeholder-driven taxonomy redesign is required. LAILA's seven-trait rubric on Qatari high-school-adjacent data [WEB-12] is the closest available proxy for an appropriate redesign; QAES (Qatari argumentative writing, 195 essays) [WEB-3] provides additional Qatar-anchored evidence.

OO-5: Critically, the output is numeric scores, but the deployment requires natural-language feedback. This is a structural validity violation: the output construct is fundamentally different. [Q11, Q50] confirm the benchmark targets numeric attribute prediction only.

ID	Evidence
Q11	"We use the same score range as the overall score range of the essays." (p.2)
Q19	"There are 5 attributes for narrative essays, namely 1. Content: The quantity of relevant text present in the essay. 2. Organization: The way the essay is structured. 3. Word Choice: The choice and aptness of the vocabulary used in the essay." (p.2)
Q50	"These annotations can be used as a gold standard for future experiments in predicting different attribute scores." (p.5)
WEB-3	QAES annotated Qatari argumentative writing with trait-specific conventions appropriate to Qatari context (https://www.mdpi.com/2306-5729/10/9/148)
WEB-12	LAILA used annotators trained on Arab high school writing — example of appropriate annotator profile for this deployment (https://arxiv.org/abs/2512.24235)
WEB-22	Arwi (CAMELBERT-MSA, ZAEUBUC) demonstrates the kind of feedback-generation output form the deployment requires; QWK is insufficient to evaluate such systems (https://arxiv.org/html/2504.11814v1)

Information Gaps

- No published official MOEHE rubric for Grades 10–12 Arabic writing was located in open sources [WEB-10]; LAILA serves as a proxy but is not authoritative.

Requires Expert Verification

- Definition of trait inventory and weighting for Qatar Grades 10–12 from MOEHE rubric documents
- Cross-country trait taxonomy variation across Arab national curricula

Remediation

Gap 1: Numeric ordinal score taxonomy with English-anchored traits ('word choice', 'sentence fluency'); deployment requires natural-language feedback aligned to Arabic rhetorical norms.

Recommendation: Adopt a trait taxonomy aligned to Qatari MOEHE rubrics (commission expert elicitation to fill the open-source rubric gap [WEB-10]); use LAILA's seven-trait rubric [WEB-12] as a starting point and add categories for morphological correctness, classical rhetorical eloquence (ةغالب), and dialect intrusion; redefine output as natural-language revision suggestions with rationale.

ID	Evidence
WEB-10	Qatar MOE Arabic Language curriculum framework anchors Grades 1–12 writing standards (http://www.sec.gov.qa/En/Education/Pages/GradeAndSubject.aspx)
WEB-12	LAILA used annotators trained on Arab high school writing — example of appropriate annotator profile for this deployment (https://arxiv.org/abs/2512.24235)